Secure Banking Application and Password Manager

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Declaration

This report has been prepared on the basis of my own work. Where other published and unpublished source materials have been used, these have been acknowledged.

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Table of Contents

Abstract……...4

Project Specification ……..... 4

Chapter 1: Introduction…….....5

Chapter 2: Literature Review……..... 6

Chapter 3: System Architecture……...11

3.1 Algorithms Used……12

Chapter 4: Module Explanation……...13

4.1 Secure Banking……...13

4.2 Password Manager……..13

Chapter 5: Results and Discussions……14

5.1 Output Screenshots……....14

5.2 Source Code…….....19

Chapter 6: Conclusion…….....25

References……..25

Abstract

In today's world online banking service is the most popular. It enables transactions to be completed quickly and easily. As the number of online transactions grows, so does the number of fraudulent transactions caused by numerous thefts. Fraudulent transactions coexist with legitimate transactions in real life. Simple pattern matching algorithms are frequently insufficient to effectively detect those forgeries. Our work focuses on detecting a fraudulent bank transaction. We will create a Secure Banking Application and a Password Manager for this project. Password Manager will be supplied to users, allowing them to have a free password manager without having to subscribe to other password managers. Secure banking application will be built by assessing if online transactions are fraudulent or non fraudulent.

Project Specification

Secure Banking Application:-We are making a secure banking application by testing whether the online transactions are fraudulent or non fraudulent.

Password Manager:-To provide a free password manager to users around the world. This will allow users to have a free password manager and they won't have to subscribe to other password managers.

1. Introduction

The financial industry is now dealing with a serious fraud problem. The issue is worldwide, and no country is completely safe. Fraudsters have mastered the art of hijacking web sessions, stealing client passwords and using malware to steal payments from unwitting account holders. According to the 2018 Global Fraud Report, 91 percent of digital marketplace customers use smartphones, with 88 percent of those using them for personal banking, and 72 percent cite fraud as a growing concern over the past twelve months, with 63 percent reporting higher levels of fraudulent losses over the same period. The development of a comprehensive system to detect fraudulent transactions is motivated by research in the field of countering various types of online frauds. As a result of this effort, we are able to distinguish between a fraudulent and a legitimate transaction.

Passwords have proven to be one of the most dependable methods for securing our data and authenticating entities. With the proliferation of online websites and our increasing reliance on them, it is no longer viable to create unique passwords for each website and then remember them. As a result, the second element of our project is a password manager that will collect passwords from the website where the user enters his login information and encrypt them using a master password. The user will be able to access all of his passwords with only one master password. Users have the option of manually entering their login information.

2. Literature Review –

1. **Renu et al and K. Ram Kalyani,2017,IPASJ International Journal of Computer Science,"Credit Card Fraud Detection in Internet using K-Nearest Neighbor Algorithm"**

   Description -
   The author of this study conducts a study of current credit card fraud detection and telephony fraud strategies. The purpose of this paper is to offer a complete overview of various fraud detection approaches.

   Limitations -
   The KNN technique for detecting outliers in data streams is given in this study. The number of scans is reduced to only one with this technique. Experiments on both synthetic and actual data sets have demonstrated that the suggested strategy is efficient and effective. It also informs the cardholder and the merchant of the transaction status at the end of the transaction procedure. Finally, it will send the user an email if a fraudulent transaction is attempted. The goal is to use an online updating methodology to find outliers in a vast amount of data.

2. **S. Praveen Kumar, A. Sahithi Choudary,2019,International Journal of Innovative Technology and Exploring Engineering (IJITEE),"An Innovative Model Based Approach for Credit Card Fraud Detection Using Predictive Analysis and Logical Regression"**

   Description -
   This study outlines a five-step strategy that is extremely effective in decreasing the fraud risks that banks face. The results show that the suggested approach, which incorporates predictive analysis, logistic regression, outlier modelling, custom rule management, and global profiling methods, is extremely successful and efficient in resolving the problem.

   Limitations -
   In this paper here is suggested an outlier model in this research, which entails developing a credit card fraud analysis model. This includes business analysis, data research, data adjustments, modelling, model verifications, and more, and it also offers us an idea of how the model will be implemented.

3. **Adit Chitnawis, Asim Ubare, Namrata Shahasane, 4Prof. Siddhesh Khanvilkar, 2019, International Journal of Applied Decision Science,"Fraud perception and prohibition for reliable Transaction of cards using data mining"**

   Description -
   The author presents a system that identifies fraud while a transaction is in progress, as well as detects fraud while a transaction is in progress.

   Limitations -
   In this paper, they presented a system in which customers who conduct transactions with cards are awarded a specific pattern on each transaction. This pattern comprises the user's location as well as the bank's transaction limit as selected by the user when

creating a bank account. The system also enables for the detection of suspicious transactions, which are promptly paused and verified, resulting in the transaction being banned if an authenticate acknowledgement is not received from the user within the specified time frame

4. **Suraj Patil, Varsha Nemade and PiyushKumar Soni,2018,International Conference on Computational Intelligence and Data Science (ICCIDS)," Predictive Modelling For Credit Card Fraud Detection Using Data Analytics"**

   Description -
   The author has presented a robust framework for processing massive amounts of data; the framework's capability may be enhanced to retrieve real-time data from a variety of sources. The data is then utilised to create a robust analytical model.

   Limitations -
   Here, they used three distinct analytical methodologies in this work to increase the analytical accuracy of fraud prediction. These analytical models are performed on a credit card dataset, and their correctness is assessed using a confusion matrix. A random forest decision tree outperforms the other two models in terms of accuracy, precision, and recall. The only issue with random forest is that as the amount of data grows, the trees in memory get overfit.

5. **Devi Meenakshi. , Janani , Gayathri and Mrs. Indira,2019,International Research Journal of Engineering and Technology (IRJET)," Credit card fraud detection using Random Forest Algorithm"**

   Description -
   The author suggested a method in which the credit card dataset is classified using a random forest algorithm.

   Limitations -
   In this paper, a random portion of the training set is sampled to train each individual tree, and then a decision tree is formed, with each node splitting on a feature chosen at random from the whole feature set. Random forest training is incredibly quick, even for big data sets with numerous characteristics and data instances, because each tree is trained independently of the others. It has been discovered that the Random Forest approach provides a decent approximation of the generalisation error and is resistant to overfitting.

6. **Vaishnave Jonnalagadda,Priya Gupta and Eesita Sen,2019,International Journal of Advance Research, Ideas and Innovations in Technology,"Credit card fraud detection using Random Forest Algorithm"**

   Description -
   To detect fraudulent credit card activity in online financial transactions, the author suggested a machine learning approach. Due to the large volume of data and its complexity, manually analysing false transactions is impossible. Machine Learning might, however, make it practicable if appropriate informational characteristics are provided.

Limitations-
The complete process of identifying mobile payment fraud based on machine learning, supervised and unsupervised methods to identify fraud and handle vast volumes of financial data was suggested in this study. Furthermore, their method used a sample and feature selection technique to handle vast amounts of transaction data quickly and accurately, resulting in high accuracy in mobile payment identification. Our suggested model is validated using the F-measure and ROC curves.

7. **Ambarish Karole1, Nitesh Saxena1, and Nicolas Christin2,2010,Information Security and Cryptology-ICISC," A Comparative Usability Evaluation of Traditional Password Managers"**

Description -
This paper has offered a usability comparison of three well-known traditional password managers, indicating that portable password managers outperform online password managers. If users choose the latter, they will have more confidence in handling their passwords.

Limitations-
In this study, they conducted a test, and the findings indicated that all users and non-students are grouped together when it comes to Online USB Phone, USB, and Phones. Students can also use an online USB phone, although Online and USB are in their own cluster. Given that the online PM was projected to be more user-friendly than the two portable PMs, this outcome is understandable. Non-students also thought the three PMs were more challenging and disappointing than students did. Students are significantly more technologically skilled than non-students, which explains why.

8. **Sanam Ghorbani Lyastani , Michael Schilling,, Sascha,2017,IPASJ International Journal of Computer Science, " Studying the Impact of Managers on Password Strength and Reuse"**

Description -
This paper presented a study on the impact of password managers on password strength and reuse, finding that users who rely on technological assistance for password formation have passwords that are both stronger and more unique, even when input through channels other than the password management. They also discovered that Google Chrome exacerbated the problem of password reuse.

Limitations-
They combined background data on users, particularly their password storage and creation tactics, with in-situ acquired password metrics, most notably the passwords' input methods, to generate a more full perspective of password managers in their study (e.g., manual entry, browser plugin, copy paste). They demonstrated that password managers do effect password strength and reuse using exploratory data analysis and statistical testing, including regression models.

9. **Paolo Gasti, Kasper B.Rasmussen, 2012, European Symposium on Research in Computer Security ESORICS, " On the security of Password based database Formats A Convenient Method for Securely Managing Passwords"**

   Description -
   This study is the first to do a thorough examination of the storage formats utilised by major password managers. Two realistic security models that are supposed to mimic the capabilities of real-world attackers have been developed.

   Limitations-
   The authors of this study do vulnerability evaluations in order to break the password, but some of the formats offered were found to be faulty, therefore further study was not performed.

10. **J. Alex Halderman ,Brent Waters , Edward W. Felten,2016, International Journal of Applied Decision Science, " A Convenient Method for Securely Managing Passwords"**

    Description -
    Attempts were made to make managing strong passwords more convenient. By employing password strengthening techniques, the design achieves a mix of mobility for mobile users and protection against brute force assaults.

    Limitations -
    The password change features in this section have not yet been added to our implementation

11. **Fahad Alodhyani, George Theodorakopoulos and Philipp Reinecke, 2020, School of Computer Science and Informatics, Cardiff University, "Password Managers—It's All about Trust and Transparency"**

    Description -
    Using Nielsen's criteria, this article examines the user interface and usability of three cloud password managers. It also uses interview research and an online questionnaire to examine the human viewpoint on the usage and non-use of password managers in terms of four major aspects—usability, trust, transparency, and security.

    Limitations -
    The usability test only looked at LastPass, but they also looked at three other cloud-based password managers because they all offer a lot of features, functionalities, and user interfaces to test.

12. **Kalpesh Vishwakarma,2020, International Research Journal of Engineering and Technology (IRJET), "Credit Card Fraudulent Detection using Machine Learning"**

Description -
The performance of K-nearest neighbours, Decision Tree, Logistic regression, and Random forest for detecting credit card fraud is examined in this research. The work is developed in Python and uses four distinct machine learning classification techniques. The accuracy, f1-score, precision, and recall scores are used to measure the algorithm's performance.

Limitations -
Although the accuracy of each method was excellent, when other assessment criteria were included, it was discovered that the Random forest classifier outperformed the KNN, Logistic regression, and even the Decision tree.

3. System Architecture –

(i) For Secure Banking –

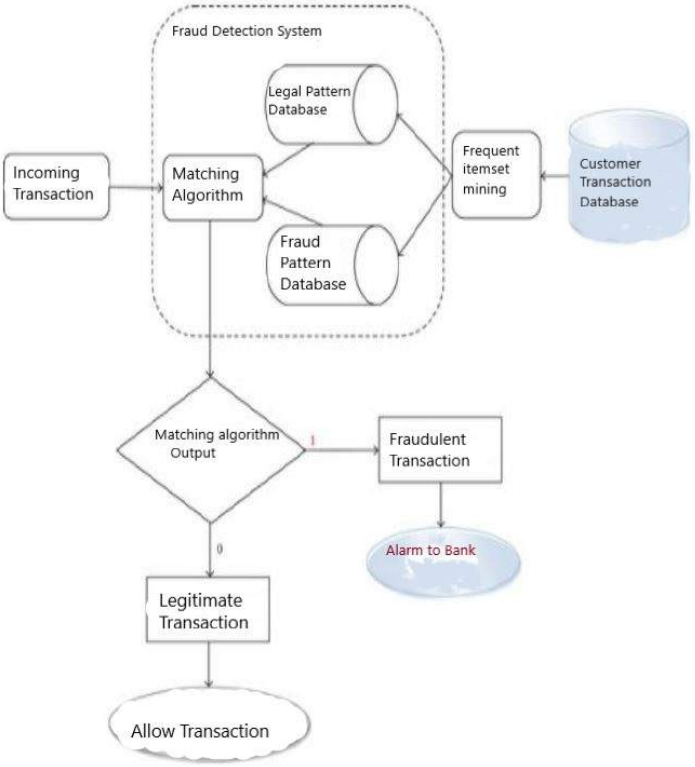


Figure 1 (Secure Banking architecture)

(ii) For Password Manager –

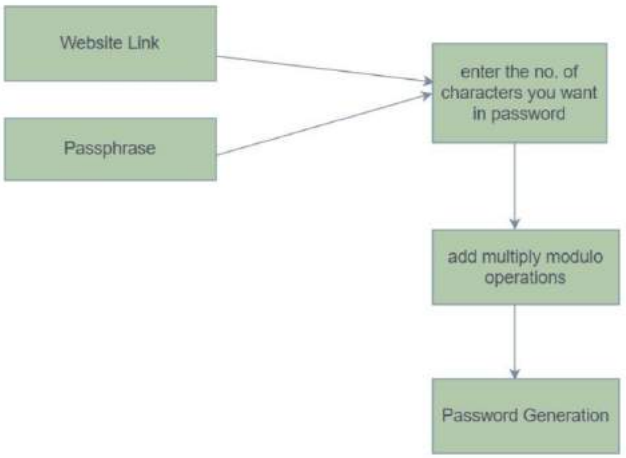


Figure 2 (Password Manager Architecture)

3.1 Algorithms Used -

Secure Banking Applications:

a) K Nearest Neighbor Algorithm: KNN is one of the most straightforward AI calculations and is utilized in a wide exhibit of establishments. It is non-parametric i.e. it implies that it doesn't make any presumptions about the hidden information and a languid learning calculation. All in all, it makes its choice dependent on the nearness to other information and focuses on paying little mind to what highlights the mathematical qualities presented. Being a sluggish learning calculation suggests that there is practically zero preparation stage. In this manner, we can quickly group new information focuses as they introduce themselves.

b) Logistic Regression: Logistic regression is a statistical method for predicting binary classes. The result or target variable is dichotomous in nature. Dichotomous methods there are just two potential classes. For instance, it tends to be utilized for malignant growth discovery issues. It figures the likelihood of an occasion occurrence.It is an uncommon instance of straight relapse where the objective variable is absolute in nature. It utilizes a log of chances as the reliant variable. Calculated Regression predicts the likelihood of event of a twofold occasion using a logit work.

c) Random Forest Classification: It's a learning method based on ensemble trees. The Random Forest Classifier is a collection of decision trees derived from a portion of the training data that has been chosen at random. Ensemble algorithms integrate more than one approach for classifying objects, whether they are of the same or different type. For instance, using Naive Bayes, SVM, and Decision Tree to make predictions and then voting on the final class for the test object.

Password Manager:

The Stanford Javascript Crypto Library (hosted on GitHub) is a project by the Stanford Computer Security Lab to build a secure, powerful, fast, small, easy-to-use, cross-browser **library for cryptography in Javascript.**

SJCL is secure. 256SHS hash function uses the industrial **AES algorithm** at 128, 192 or 256 bits.The default boundaries are reasonable: SJCL makes your passwords strong by 1000 times and it becomes secure, and it validates each message.It can not be changed. We accept that SJCL gives the best security which is for all intents and purposes accessible in Javascript.

4. Module Explanation –

4.1 Secure Banking -

- The dataset is being processed.
- Identifying the likelihood and kind of a fraudulent transaction.
- Plotting the relationship between the characteristics of fraudulent and legitimate transactions.
- Creating the train and divides for testing.
- Data classification utilising various algorithms such as KNN, Random Forest, and Logical Regression, with accuracy and efficiency in identifying frauds calculated.
- On the basis of accuracies, recall, and precision, a comparative study of all Classification algorithms is performed.

Data dictionary for secure banking

- step - represents a unit of time in the real world. There are a total of 744 stages in this simulation (30 days).
- CASH-IN, CASH-OUT, DEBIT, PAYMENT, and TRANSFER are the different types of transactions.
- amount - the transaction's total value in local currency.
- nameOrig - the customer who initiated the transaction;
- oldbalanceOrg - the initial balance before the transaction;
- newbalanceOrig - the new balance after the transaction;
- nameDest - the customer who is the transaction's recipient;
- oldbalanceDest - the initial balance recipient before the transaction; newbalanceDest - the new balance recipient after the transaction;
- isFraud - This represents the fraudulent agents' transactions within the simulation. The specialists' deceptive behaviour in this dataset points to profit by seizing possession of the clients' records and attempting to delete the assets by switching to another record and then exiting the framework.
- isFlaggedFraud - The business model aims to control massive transfers from one record to another and banners illicit endeavors. An unlawful endeavor in this dataset is an endeavor to move more than 200.000 in a solitary exchange.

4.2 Password Manager –

- HTML module is used to create a webpage that accepts the master password and also stores, displays passwords of different websites.
- CSS is used to format the webpage,that controls the spacing and leading of text.
- Javascript makes our web page dynamic, making it interactive and user friendly.

5. Results and Discussions –

5.1 Output Screenshots -

For Secure Banking :

```
1047433
1142
Percentage frauds: 0.0010902845337124188
```



Figure 5.1.1. - Plotting number of frauds

Types of transactions that have Frauds: ['TRANSFER', 'CASH_OUT']

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
2	1	TRANSFER	181.00	C1305486145	181.00	0.00	C553264065	0.00	0.00	1	0
3	1	CASH_OUT	181.00	C840083671	181.00	0.00	C38997010	21182.00	0.00	1	0
15	1	CASH_OUT	229133.94	C905080434	15325.00	0.00	C476402209	5083.00	51513.44	0	0
19	1	TRANSFER	215310.30	C1670993182	705.00	0.00	C1100439041	22425.00	0.00	0	0
24	1	TRANSFER	311685.89	C1984094095	10835.00	0.00	C932583850	6267.00	2719172.89	0	0
...	...	...	...	...	...	...	...	...	...	...	...
1048553	95	CASH_OUT	103391.93	C2021893664	107623.00	4231.07	C178178755	5752648.68	5856040.61	0	0
1048558	95	CASH_OUT	26535.41	C1213094114	104187.00	77651.59	C216374600	205460.26	231995.66	0	0
1048565	95	TRANSFER	132387.24	C1654402840	15956.51	0.00	C1878219072	631284.08	763671.32	0	0
1048567	95	CASH_OUT	279674.05	C990252469	18002.85	0.00	C574439165	1847488.28	2127162.32	0	0
1048570	95	CASH_OUT	132557.35	C1179511630	479803.00	347245.65	C435674507	484329.37	616886.72	0	0

460394 rows × 11 columns

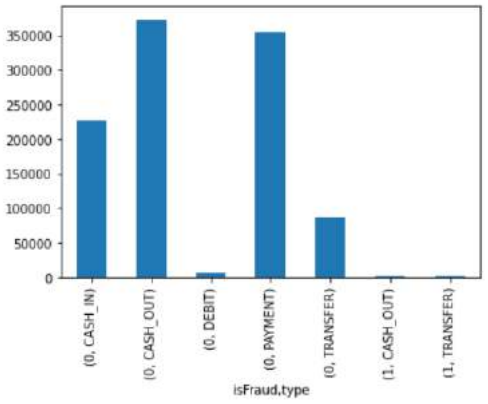


Figure 5.1.2 - Distribution of the fraud transactions in the type of data

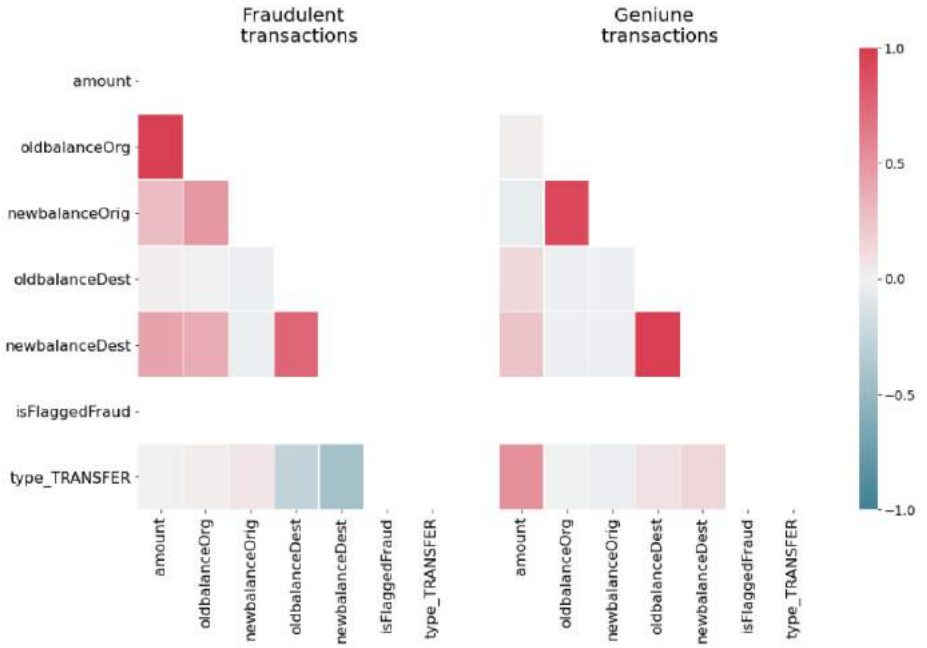


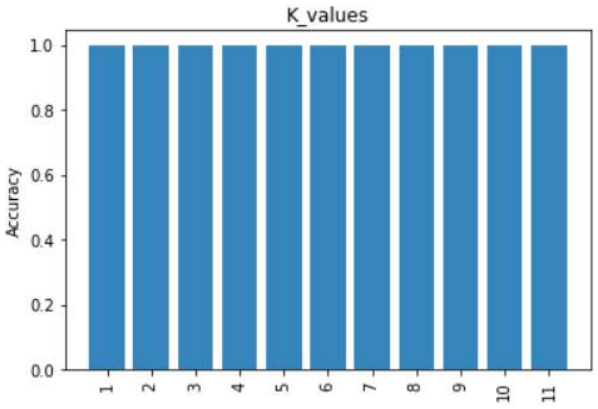
Figure 5.1.3 - Plotting the correlation between the features for fraud and genuine transactions

```
Logistic Regression
Accuracy: 99.8315024583528 %
Recall: 72.9381443298969 %
Precision: 65.2073732718894 %
F1 score: 68.85644768856447 %
```

Figure 5.1.4 - Classification using Logistic regression

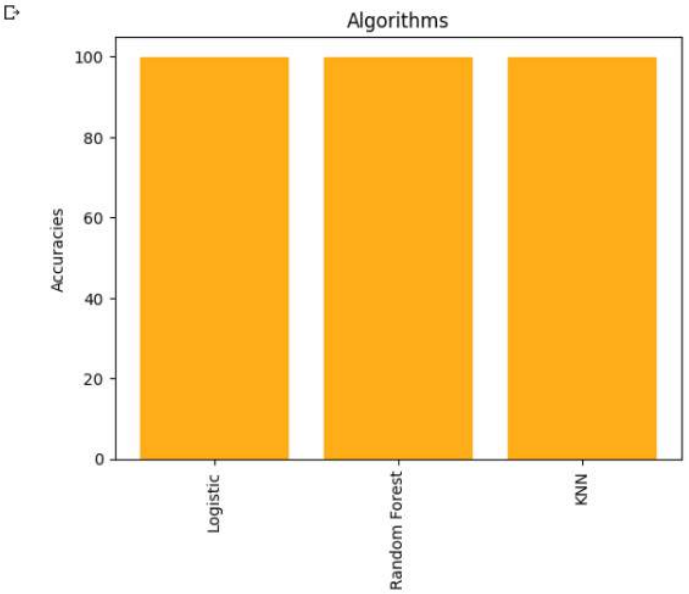
```
Random Forest Method:
Accuracy: 99.90851110043376 %
Recall: 67.26804123711341 %
Precision: 95.6043956043956 %
F1 score: 78.9712556732224 %
```

Figure 5.1.5 Classification using Random Forest method




```
KNN Method without applying SMOTE-
Accuracy: 99.8802087789852 %
Recall: 60.567010309278345 %
Precision: 89.01515151515152 %
F1 score: 72.08588957055214 %
```

Figure 5.1.6 - Classification using KNN classifier



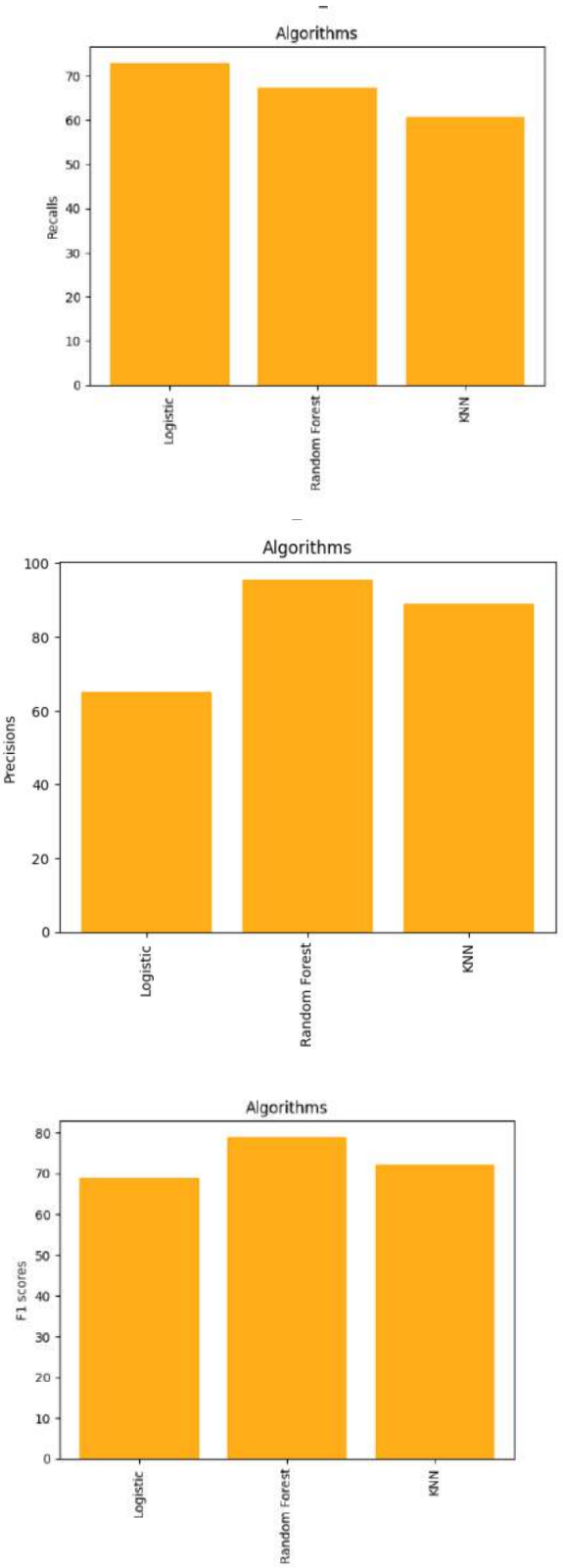


Figure 5.1.7 - Plotting all required metrics to compare the algorithms

For Password Manager:

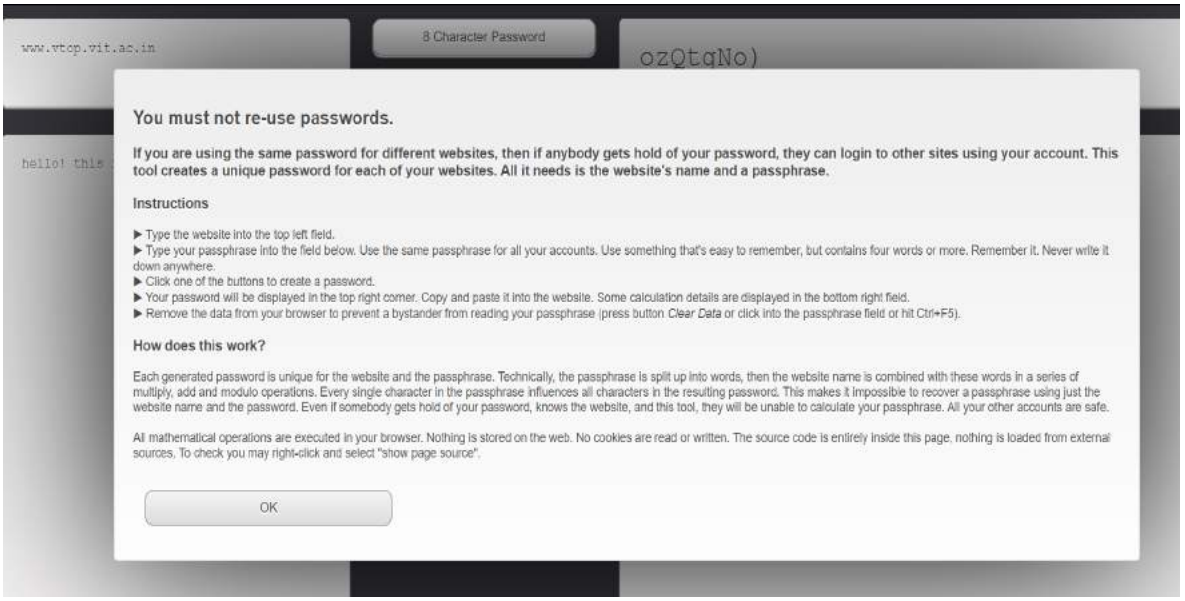


Figure 5.1.8 - (Front end part of the website)

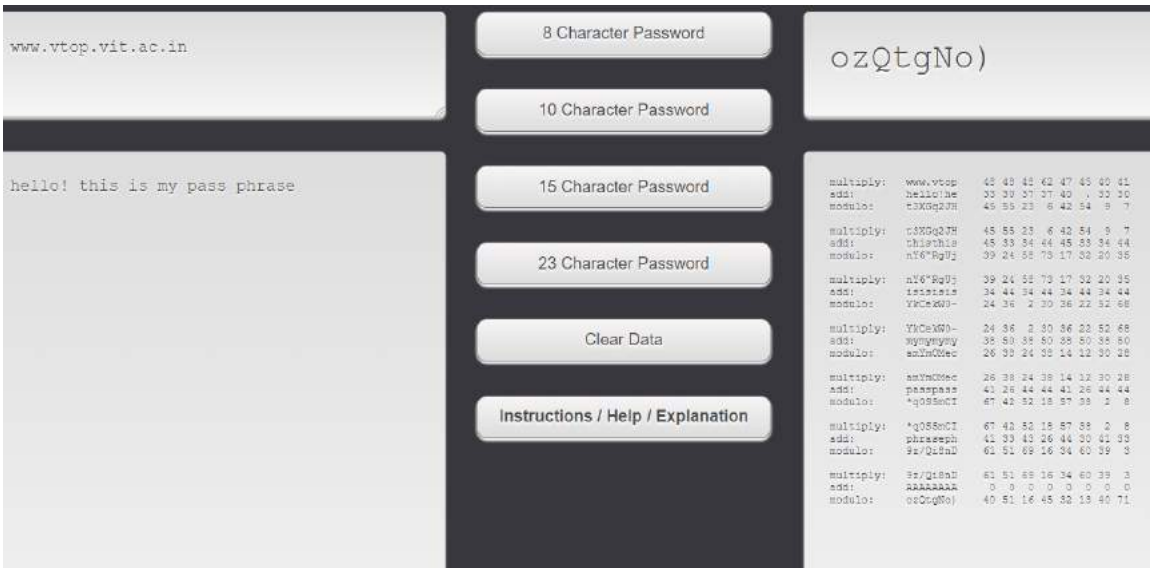


Figure 5.1.9 (Basic functioning part of the webpage)

5.2 Source Code -

Secure Banking Application Code

+ Code + Text

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving testfile.csv to testfile (1).csv
```

```
#importing required packages
import io
import warnings; warnings.simplefilter('ignore')
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import recall_score
from sklearn.metrics import precision_score
from sklearn.metrics import f1_score
from imblearn.over_sampling import SMOTE

# uploading dataset
df2 = df2 = pd.read_csv(io.BytesIO(uploaded['testfile.csv']))
df2
```

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
0	1	PAYMENT	9839.64	C1231006815	170136.00	160296.36	M1979787155	0.00	0.00	0	0
1	1	PAYMENT	1864.28	C1666544295	21249.00	19384.72	M2044282225	0.00	0.00	0	0
2	1	TRANSFER	181.00	C1305486145	181.00	0.00	C553264065	0.00	0.00	1	0
3	1	CASH_OUT	181.00	C840083671	181.00	0.00	C38997010	21182.00	0.00	1	0
4	1	PAYMENT	11668.14	C2048537720	41554.00	29885.86	M1230701703	0.00	0.00	0	0
...	...	...	...	...	...	...	...	...	...	...	...
1048570	95	CASH_OUT	132557.35	C1179511630	479803.00	347245.65	C435674507	484329.37	616886.72	0	0
1048571	95	PAYMENT	9917.36	C1956161225	90545.00	80627.64	M668364942	0.00	0.00	0	0
1048572	95	PAYMENT	14140.05	C2037964975	20545.00	6404.95	M1355182933	0.00	0.00	0	0
1048573	95	PAYMENT	10020.05	C1633237354	90605.00	80584.95	M1964992463	0.00	0.00	0	0
1048574	95	PAYMENT	11450.03	C1264356443	80584.95	69134.92	M677577406	0.00	0.00	0	0

1048575 rows × 11 columns

```
[ ] #plotting the number of frauds and non frauds
    %matplotlib inline
    import matplotlib.pyplot as plt
    frauds = list(df2['isFraud'])
    frauds

    # print(frauds)
    count_0 = 0
    count_1 = 0
    for i in frauds:
      if (i==0):
        count_0 += 1
      else:
        count_1 += 1
    print(count_0)
    print(count_1)
    print('Percentage frauds:',count_1/count_0)

    objects = ('Frauds', 'Non-Frauds')
    y_pos = np.arange(len(objects))

    #The number of frauds are too less to be depicted in the plot, so the count of non-frauds has been divided by 100 for better clarity
    performance = [count_1,count_0/100]
    plt.bar(y_pos, performance, align='center', alpha=0.3)
    plt.xticks(y_pos, objects)
    plt.ylabel('Counts')
    plt.title('Frauds')
    plt.show()
```

```
[ ]  # the below data and plot shows the distribution of the fraud transactions in the type of data

     # types of fraud transactions
     fraud_trans = list(df2.loc[df2.isFraud==1].type.drop_duplicates().values)
     print('Types of transactions that have Frauds: ',fraud_trans)

     # plot for the same
     fraud_count = df2.groupby(['isFraud','type']).size().plot(kind='bar')

     # removing the data which do not have fraudalant transactions
     trans_data_new = df2.loc[(df2['type'].isin(['TRANSFER','CASH_OUT'])),:]
     trans_data_new
```

```
#Plotting the correlation between the features for fraud and genuine transactions
X = trans_data_new.drop(columns=['isFraud'])

# # removing the columns with low co-relation coefficient
X = X.drop(columns=['nameOrig','nameDest','step'])

# # Handling the categorical column using dummies in python
X = pd.get_dummies(X,columns=['type'],drop_first=True)

# # getting the dependent variable
y = pd.DataFrame(trans_data_new['isFraud'])

import seaborn as sns
Xfraud = X.loc[y['isFraud']==1] # update Xfraud & XnonFraud with cleaned data
Xnonfraud = X.loc[y['isFraud']==0]

correlationFraud = Xfraud.loc[:, X.columns != 'step'].corr()
mask = np.zeros_like(correlationFraud)
indices = np.triu_indices_from(correlationFraud)
mask[indices] = True

grid_kws = {"width_ratios": (.9, .9, .05), "wspace": 0.2}
f, (ax1, ax2, cbar_ax) = plt.subplots(1, 3, gridspec_kw=grid_kws, \
                                      figsize = (14, 9))

cmap = sns.diverging_palette(220, 8, as_cmap=True)
ax1 =sns.heatmap(correlationFraud, ax = ax1, vmin = -1, vmax = 1, \
    cmap = cmap, square = False, linewidths = 0.5, mask = mask, cbar = False)
ax1.set_xticklabels(ax1.get_xticklabels(), size = 16);
ax1.set_yticklabels(ax1.get_yticklabels(), size = 16);
ax1.set_title('Fraudulent \n transactions', size = 20)

correlationnonFraud = Xnonfraud.loc[:, X.columns != 'step'].corr()
ax2 = sns.heatmap(correlationnonFraud, vmin = -1, vmax = 1, cmap = cmap, \
 ax = ax2, square = False, linewidths = 0.5, mask = mask, yticklabels = False, \
    cbar_ax = cbar_ax, cbar_kws={'orientation': 'vertical', \
                                 'ticks': [-1, -0.5, 0, 0.5, 1]})
ax2.set_xticklabels(ax2.get_xticklabels(), size = 16);
ax2.set_title('Geniune \n transactions', size = 20);
```

```
[ ]  # Snippet to create the train and test splits
     X = trans_data_new.drop(columns=['isFraud'])

     # removing the columns with low co-relation coefficient
     X = X.drop(columns=['nameOrig','nameDest','step'])

     # Handling the categorical column using dummies in python
     X = pd.get_dummies(X,columns=['type'],drop_first=True)

     # getting the dependent variable
     y = pd.DataFrame(trans_data_new['isFraud'])

     # getting the train and test data
     X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33, random_state=42)
```

```
# PLotting all required metrics to compare the algorithms
import matplotlib.pyplot as plt; plt.rcdefaults()
import matplotlib.pyplot as plt

#plotting accuracies
algos = ('Logistic','Random Forest','KNN')
accuracies = np.arange(len(algos))
performance = [lr_ac,rf_ac,knn_ac]
bars = plt.bar(accuracies, performance, align='center', alpha=0.9)
for i in range(len(bars)):
    if(i<4):
        bars[i].set_color('orange')
    else: bars[i].set_color('green')
plt.xticks(accuracies, algos,rotation='vertical')
plt.ylabel('Accuracies')
plt.title('Algorithms')
plt.show()

#plotting recall scores
algos = ('Logistic','Random Forest','KNN')
recalls = np.arange(len(algos))
performance = [lr_rs,rf_rs,knn_rs]
bars = plt.bar(recalls, performance, align='center', alpha=0.9)
for i in range(len(bars)):
    if(i<4):
        bars[i].set_color('orange')
    else: bars[i].set_color('green')
plt.xticks(recalls, algos,rotation='vertical')
plt.ylabel('Recalls')
plt.title('Algorithms')
plt.show()
```

```
#plotting precisions
algos = ('Logistic','Random Forest','KNN')
precisions = np.arange(len(algos))
performance = [lr_ps,rf_ps,knn_ps]
bars = plt.bar(precisions, performance, align='center', alpha=0.9)
for i in range(len(bars)):
    if(i<4):
        bars[i].set_color('orange')
    else: bars[i].set_color('green')
plt.xticks(precisions, algos,rotation='vertical')
plt.ylabel('Precisions')
plt.title('Algorithms')
plt.show()

#plotting F1 scores
algos = ('Logistic','Random Forest','KNN')
fscores = np.arange(len(algos))
performance = [lr_fs,rf_fs,knn_fs]
bars = plt.bar(fscores, performance, align='center', alpha=0.9)
for i in range(len(bars)):
    if(i<4):
        bars[i].set_color('orange')
    else: bars[i].set_color('green')
plt.xticks(fscores, algos,rotation='vertical')
plt.ylabel('F1 scores')
plt.title('Algorithms')
plt.show()
```

Password Manager:

password-manager.html 1 ×

C > Users > HP > Downloads > Nasscom Project > password-manager.html > ...

```
<!DOCTYPE html>
<html>


    <head>
        <title>Password Manager</title>


        <style>

            html, body {
                background-color: rgb(55, 55, 61);
                width: 100vw;
                height: 100vh;
                margin: 0;
                padding: 0;
                border: 0;
                font-family: Sans-Serif;
                overflow: hidden;
            }

            button {
                position: relative;
                float: left;
                clear: left;
                margin: 1vw;
                width: 100%;
                border: 1px solid #777272;
                -webkit-box-shadow: #9fafaf 0px 2px 2px ;
                -moz-box-shadow: #FEFFFF 0px 2px 2px ;
                box-shadow: #FEFFFF 0px 2px 2px ;
                -webkit-border-radius: 10px;
                -moz-border-radius: 10px;
                border-radius: 10px;
                padding: 10px 10px 10px 10px;
                text-shadow: 0px 2px 2px rgba(255,255,255,1);
                color: #4A4A4A;
                background-color: #8a717d;
                background-image: -webkit-gradient(linear, left top, left bottom, from(#F7F5F6), to(#DDDDDD));
                background-image: -webkit-linear-gradient(top, #F7F5F6, #DDDDDD);
                background-image: -moz-linear-gradient(top, #F7F5F6, #DDDDDD);
                background-image: -ms-linear-gradient(top, #F7F5F6, #DDDDDD);
                background-image: -o-linear-gradient(top, #F7F5F6, #DDDDDD);
                background-image: linear-gradient(to bottom, #F7F5F6, #DDDDDD);
                filter:progid:DXImageTransform.Microsoft.gradient(GradientType=0,startColorstr=#F7F5F6, endColorstr=#DDDDDD);
                white-space: normal;
                font-size: 1vw;
            }

            button:hover{
                border:1px solid #ADADAD;
                background-color: #E0E0E0;
                background-image: -webkit-gradient(linear, left top, left bottom, from(#E0E0E0), to(#BDBBBC));
                background-image: -webkit-linear-gradient(top, #E0E0E0, #BDBBBC);
                background-image: -moz-linear-gradient(top, #E0E0E0, #BDBBBC);
                background-image: -ms-linear-gradient(top, #E0E0E0, #BDBBBC);
                background-image: -o-linear-gradient(top, #E0E0E0, #BDBBBC);
                background-image: linear-gradient(to bottom, #E0E0E0, #BDBBBC);
                filter:progid:DXImageTransform.Microsoft.gradient(GradientType=0,startColorstr=#E0E0E0, endColorstr=#BDBBBC);
            }

            button:active{
                border:1px solid #ADADAD;
                background-color: #E0E0E0;
                background-image: -webkit-gradient(linear, left top, left bottom, from(#BDBBBC), to(#E0E0E0));
                background-image: -webkit-linear-gradient(top, #BDBB8C, #E0E0E0);
                background-image: -moz-linear-gradient(top, #BDBBBC, #E0E0E0);
                background-image: -ms-linear-gradient(top, #BDBB8C, #E0E0E0);
                background-image: -o-linear-gradient(top, #BDBBBC, #E0E0E0);
                background-image: linear-gradient(to bottom, #BDBBBC, #E0E0E0);
                filter:progid:DXImageTransform.Microsoft.gradient(GradientType=0,startColorstr=#BDBBBC, endColorstr=#E0E0E0);
            }

            div {
                text-shadow: 0px 2px 2px rgb(216, 191, 191);
                color: #202020;
            }

            textarea {
                font-family: mono, courier, fixed, serif;
                position: relative;
                float: left;
                clear: left;
                border: 0.1px solid #CCCCCC;
                padding: 1.5vw;
                margin: 1vw;
                width: 100%;
                font-size: 1.5vh;
```

```
            -webkit-box-shadow: #CCCCCC 0px 2px 2px ;
            -moz-box-shadow: #CCCCCC 0px 2px 2px ;
            box-shadow: #CCCCCC 0px 2px 2px ;
            -webkit-border-radius: 5px;
            -moz-border-radius: 5px;
            border-radius: 5px;
            text-shadow: 0px 2px 2px rgba(255,255,255,1);
            color: #4A4A4A;
            background-color: #F7F5F6;
            background-image: -webkit-gradient(linear, left top, left bottom, from(#DDDDDD), to(#F7F5F6));
            background-image: -webkit-linear-gradient(top, #DDDDDD, #F7F5F6);
            background-image: -moz-linear-gradient(top, #DDDDDD, #F7F5F6);
            background-image: -ms-linear-gradient(top, #DDDDDD, #F7F5F6);
            background-image: -o-linear-gradient(top, #DDDDDD, #F7F5F6);
            background-image: linear-gradient(to bottom, #DDDDDD, #F7F5F6);
            filter:progid:DXImageTransform.Microsoft.gradient(GradientType=0,startColorstr=#DDDDDD, endColorstr=#F7F5F6);
        }


        div.block {
            margin: 0;
            padding: 0;
            font-family: verdana, arial, sansserif;
            position: relative;
            float: left;
            height: 100vh;
            white-space: normal;
            vertical-align: top;
            text-align: middle;
        }
```

```
        div.help {
            position: absolute;
            left: 10vw;
            top: 10vh;
            width: 80vw;
            display:none;
            border: 0.1px solid #CCCCCC;
            padding: 1.5vw;
            margin: 0;
            font-size: 2vh;
            -webkit-box-shadow: #000 0px 0px 400px ;
            -moz-box-shadow: #000 0px 0px 400px ;
            box-shadow: #000 0px 0px 400px ;
            -webkit-border-radius: 5px;
            -moz-border-radius: 5px;
            border-radius: 5px;
            text-shadow: 0px 2px 2px rgba(255,255,255,1);
            color: #4A4A4A;
            background-color: #F7F5F6;
            background-image: -webkit-gradient(linear, left top, left bottom, from(#EEEEEE), to(#FEFEFE));
            background-image: -webkit-linear-gradient(top, #EEEEEE, #FEFEFE);
            background-image: -moz-linear-gradient(top, #EEEEEE, #FEFEFE);
            background-image: -ms-linear-gradient(top, #EEEEEE, #FEFEFE);
            background-image: -o-linear-gradient(top, #EEEEEE, #FEFEFE);
            background-image: linear-gradient(to bottom, #EEEEEE, #FEFEFE);
            filter:progid:DXImageTransform.Microsoft.gradient(GradientType=0,startColorstr=#EEEEEE, endColorstr=#FEFEFE);
        }

        </style>

```

```
    <script type="text/javascript">
        <!-
        function Dict74()
        {
            this.dict = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789.,:;+*-/()'\"";
        }
        Dict74.prototype.code = function(word)
        {
            result = [];
            for (var i=0; i<word.length; i++)
                result.push( this.dict.indexOf( word[i] ) );
            return result;
        }
        Dict74.prototype.decode = function(array)
        {
            result = "";
            for (var i=0; i<array.length; i++)
                result += this.dict[array[i]];
            return result;
        }
        function toString(array)
        {
            result = "";
            for (var i=0; i<array.length; i++)
                if (array[i] < 0)
                    result += "  .";
                else if (array[i] < 10)
                    result += "  " + array[i].toString();
                else
                    result += " " + array[i].toString();
```

```
                return result;
            }
            var _dict74 = new Dict74();
            function expand(word, letters)
            {
                result = "";
                for (i=0, w=0; i<letters; i++, w++) {
                    if (w >= word.length)
                        w = 0;
                    result += word[w];
                }
                return result;
            }
            function combine(first, second, n)
            {
                result = [];
                for (var i=0; i<n; i++) {
                    char = second[i];
                    for (var j=0; j<n; j++) {
                        char += first[j] * Math.abs(i-j);
                    }
                    char = Math.abs(char) % 74;
                    result.push(char);
                }
                return result;
            }
            function calculate(n)
            {
```

```
                url = document.getElementById("url").value.trim();
                url = url.replace(/ /g,'');
                if (url==='')
                    url = ".";
                words = document.getElementById("phrase").value.trim();
                if (words === '')
                    words = ". . . . .";
                words = words.split(' ');
                words = [url].concat(words).concat("A");


                for (var i=0; i<words.length; i++)
                    words[i] = expand(words[i], n);


                string = "";
                combineStr = words[0];
                combineArr = _dict74.code(combineStr);
                for (var w=1; w<words.length; w++) {
                    wordStr = words[w];
                    wordArr = _dict74.code(wordStr);
                    string += "multiply:    " + combineStr + "   " + toString(combineArr) + "\n";
                    string += "add:         " + wordStr + "    " + toString(wordArr) + "\n";

                    combineArr = combine(combineArr, wordArr, n);
                    combineStr = _dict74.decode(combineArr);
                    string += "modulo:      " + combineStr + "   " + toString(combineArr) + "\n";
                    string += "\n";
                }
                document.getElementById("progress").value = string;
```

```
                document.getElementById("password").value = combineStr;
                document.getElementById("password").focus();
                document.getElementById("password").select();
            }
            function clearData(id)
            {
                if (id==='all') {
                    clearData("url");
                    clearData("phrase");
                    clearData("progress");
                    clearData("password");
                }
                else
                    document.getElementById(id).value = "";
            }
            //-->
        </script>
    </head>
    <body>
        <div class="block" style="width:30vw;">
            <textarea id="url" wrap="hard" rows="1" style="height:7vh; font-size:1.8vw; width:25vw;" onfocus="clearData('url')">Replace thi
            <textarea id="phrase" wrap="hard" rows= "200" style="height:70vh; font-size:1.0vw; width:25vw;" onfocus="clearData('phrase')">R
        </div>
        <div class="block" style="width:20vw">
            <button onClick="calculate(8)" style="width:18vw">8 Character Password</button>
            <button onClick="calculate(10)" style="width:18vw">10 Character Password</button>
            <button onClick="calculate(15)" style="width:18vw">15 Character Password</button>
            <button onClick="calculate(23)" style="width:18vw">23 Character Password</button>
            <button onClick="clearData('all')" style="width:18vw">Clear Data</button>
            <button onclick="document.getElementById('help').style.display='block';" style="font-weight: bold; width:18vw">Instructions / H
        </div>
```

```
            <button onclick="document.getElementById('help').style.display='block';" style="font-weight: bold; width:18vw">Instructions / H
        </div>
        <div class="block" style="width:50vw">
            <textarea id="password" wrap="hard" rows="1" style="height:7vh; font-size:2.1vw; width:45vw;">Your Password</textarea><br/>
            <textarea id="progress" wrap="hard" rows="200" style="height:70vh; font-size:0.65vw; width:45vw;">Calculation Details:</textarea
        </div>
        <div id="help" class="help" style="width:80vw">
            <h2>You must not re-use passwords.</h2>
            <h3>If you are using the same password for different websites, then if anybody gets hold of your password, they can login to ot

            <h3>Instructions</h3>
            &#9654; Type the website into the top left field.<br/>
            &#9654; Type your passphrase into the field below. Use the same passphrase for all your accounts. Use something that's easy to
            &#9654; Click one of the buttons to create a password.<br/>
            &#9654; Your password will be displayed in the top right corner. Copy and paste it into the website. Some calculation details a
            &#9654; Remove the data from your browser to prevent a bystander from reading your passphrase (press button <i>Clear Data</i> o

            <h3>How does this work?</h3>
            Each generated password is unique for the website and the passphrase. Technically, the passphrase is split up into words, then
            All mathematical operations are executed in your browser. Nothing is stored on the web. No cookies are read or written. The sou
            <button onclick="document.getElementById('help').style.display='none';" style="width: 20vw;">OK</button>
            <br/><br/>
        </div>
    </body>

</html>
```

6. Conclusion

In this project, we are trying to build a predictive model for predicting fraud and a simple application to simulate the model running on unseen data and also a Password manager in which we set a master password and extract all other passwords with that single master password. This will help us in generating and saving strong, unique passwords when we sign up to new websites. That means whenever we go to a website or app, we can pull up our password manager, copy your password, paste it into the login box, and we're in. For better visualization and understanding, plots have been used. From the metrics above, we see that the model is very promising to be productionized.

7. References

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